Spotting Yellow Leaf Disease from Space

Satellite-based mapping and early detection of arecanut Yellow Leaf Disease (YLD) | Sringeri & Sulya taluks, Karnataka

INTERNSHIP PROJECT
September – December

≈50%
YIELD LOSS

Yellow Leaf Disease (phytoplasma) can cut arecanut yield by about half within three years. Sringeri and Sullia growers have been hit hard.

87–96%
INCIDENCE

YLD incidence in surveyed Sringeri and Koppa villages (2018). Field surveys check only a handful of gardens per location, so wide-area monitoring is missing.

72
MONTHLY DATASETS

Our idea: free Sentinel-2 images, 3 years × 12 months × 2 taluks, to watch every garden, catch stress early and map how the disease spreads.

OUR APPROACH IN SIX STEPS

From raw satellite images to disease maps, all in QGIS

STEP 1 — Collect

Free Sentinel-2 images of both taluks, every month for 3 years, plus rainfall and temperature records.

USES
Copernicus Browser · KSNDMC · KGIS · Google Earth Pro

GIVES
72 taluk-month datasets

STEP 2 — Clean

Remove clouds and shadows, align the layers and blend each month into one clear picture.

USES
QGIS · cloud mask (SCL band)

GIVES
Cloud-free monthly images

STEP 3 — Measure

Turn colours into numbers. 15 vegetation indices show greenness, chlorophyll and water stress.

USES
QGIS Raster Calculator

GIVES
Index maps for every month

STEP 4 — Find

Train a Random Forest classifier on Google Earth and field points to recognise arecanut, then map every garden.

USES
Random Forest classifier

GIVES
Arecanut distribution map

STEP 5 — Diagnose

Compare each garden with healthy ones, adjust for rain and drought, and flag Healthy, Stressed or Suspected YLD.

USES
Index anomalies + weather data

GIVES
YLD severity map

STEP 6 — Track

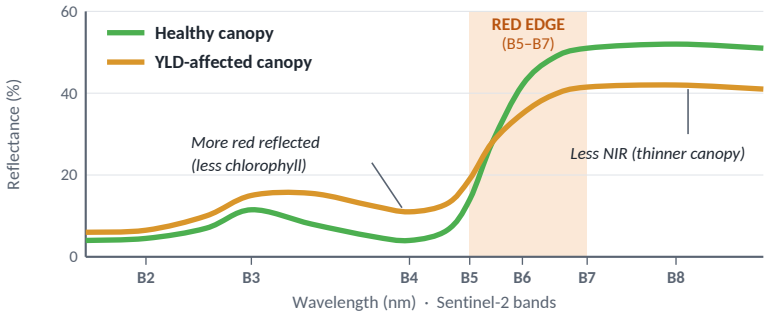
Repeat every month for 3 years to see where the disease appears and how fast it spreads.

USES
Time series · hotspot analysis

GIVES
Trend and hotspot report

CHECKED AT EVERY STEP Field GPS points and Google Earth Pro validate every map · targets: overall accuracy ≥ 85 %, Kappa ≥ 0.80 · errors feed back to improve Steps 4 and 5

Why yellowing is visible from space



Schematic. Yellow leaves reflect more red and a thinner canopy reflects less near-infrared. The red-edge bands react first, so NDRE, CIre and PSRI are core to our indexing. Measured spectra from our field plots will replace this sketch.

The result: maps that show the spread



Illustrative example. Gardens shift from healthy to stressed to suspected YLD. Our multi-temporal maps will measure this spread in hectares, by taluk and year.

- Arecanut distribution maps
- YLD severity maps
- Trend & hotspot report
- GIS-ready datasets

Project at a glance

2 taluks: Sringeri and Sulya	**3 yrs** of monthly imagery
72 taluk-month datasets	**15** vegetation indices
10 m pixel size (Sentinel-2)	**4 mo** Sep–Dec internship

Sources for context figures: Deccan Chronicle (Oct 2021); YLD incidence survey, Sringeri & Koppa (Sept 2018); YLD symptoms and yield-loss data article (PMC8551408); YLD survey, Chikkamagaluru taluks (2021–22).

The 15 Indices, in Plain English

Each index turns satellite colours into one number that answers a simple question about the canopy

STEP 3 · MEASURE
Sentinel-2 L2A surface reflectance

Greenness & biomass

Is there a healthy, dense canopy?

NDVI — PROPOSAL
Normalized Difference Vegetation Index
How green and dense is the canopy?
(B8 − B4) / (B8 + B4)

RVI — PROPOSAL
Ratio Vegetation Index
How much leaf biomass is there?
B8 / B4

SAVI — PROPOSAL
Soil-Adjusted Vegetation Index
Greenness with bare-soil glare removed
1.5 × (B8 − B4) / (B8 + B4 + 0.5)

EVI
Enhanced Vegetation Index
Greenness that does not saturate in thick canopy
2.5 × (B8 − B4) / (B8 + 6·B4 − 7.5·B2 + 1)

Chlorophyll & red edge

Is the leaf losing its green pigment?

GNDVI
Green NDVI
Early hint of falling chlorophyll (green-based)
(B8 − B3) / (B8 + B3)

NDRE — KEY FOR YLD
Normalized Difference Red Edge
Early alarm for chlorophyll loss
(B8A − B5) / (B8A + B5)

CIre — KEY FOR YLD
Chlorophyll Index − Red Edge
How much chlorophyll is in the canopy?
(B7 / B5) − 1

MTCI
MERIS Terrestrial Chlorophyll Index
Chlorophyll from the red-edge position
(B6 − B5) / (B5 − B4)

IRECI
Inverted Red-Edge Chlorophyll Index
Chlorophyll and leaf area in dense canopy
(B7 − B4) / (B5 / B6)

Yellowing & water stress

Is it turning yellow or getting thirsty?

PSRI — KEY FOR YLD
Plant Senescence Reflectance Index
Is the canopy turning yellow?
(B4 − B2) / B6

NDMI — KEY FOR YLD
Normalized Difference Moisture Index
Is it thirsty? Tells drought from disease
(B8A − B11) / (B8A + B11)

MSI
Moisture Stress Index
Water-stress level (higher means drier)
B11 / B8

Land-cover masking

Which pixels are not vegetation?

NDWI
Normalized Difference Water Index
Where is open water?
(B3 − B8) / (B3 + B8)

NDBI
Normalized Difference Built-up Index
Where are buildings and settlements?
(B11 − B8) / (B11 + B8)

BSI
Bare Soil Index
Where is bare or fallow land?
((B11+B4) − (B8+B2)) / ((B11+B4) + (B8+B2))

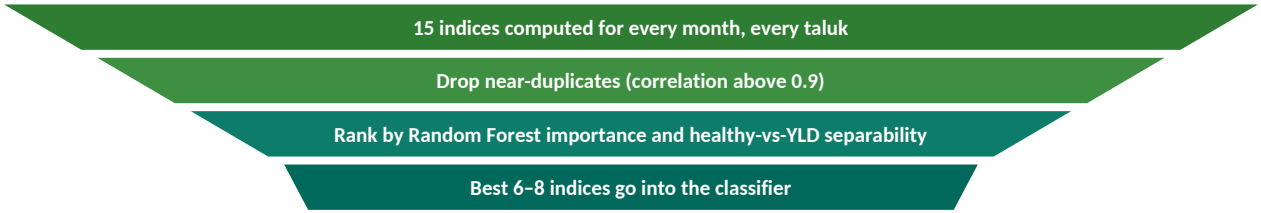
Sentinel-2 bands behind these formulas

B2	B3	B4	B5	B6
Blue	Green	Red	Red edge 1	Red edge 2
490 nm · 10 m	560 nm · 10 m	665 nm · 10 m	705 nm · 20 m	740 nm · 20 m

B7	B8	B8A	B11
Red edge 3	Near-IR	Narrow NIR	SWIR
783 nm · 20 m	842 nm · 10 m	865 nm · 20 m	1610 nm · 20 m

10 m bands give sharp detail; 20 m bands are resampled to 10 m.

HOW WE PICK THE BEST INDICES



Why this set?

- Red-edge indices and PSRI react to chlorophyll loss earlier than NDVI, so they give the earliest YLD warning.
- NDMI and MSI tell drought stress apart from disease.
- NDVI, RVI and SAVI from the proposal stay as baseline comparisons.
- The data decides the final set, not guesswork.

Work Plan, Risks and Deliverables

Four months, four clear milestones, and a plan for every known difficulty

SEPTEMBER – DECEMBER
16 weeks

SEP — Learn & collect

- Review YLD and remote-sensing literature
- Download 3 years of Sentinel-2 and KSNDMC weather
- Collect KGIS boundaries; plan field visits

◆ Milestone: Data ready

OCT — Clean & measure

- Cloud masking and monthly composites
- Compute the 15 indices
- Select the best 6–8 features

◆ Milestone: Index stack ready

NOV — Map & diagnose

- Arecanut mapping (Random Forest)
- Health and YLD classification
- Field and Google Earth checks

◆ Milestone: Arecanut and YLD maps

DEC — Track & deliver

- Multi-temporal and hotspot analysis
- Accuracy assessment
- Final report and GIS datasets

◆ Milestone: Final report

KNOWN CHALLENGES, AND HOW WE HANDLE THEM

⚠ **Monsoon clouds hide the ground (June–September)**	✓ Monthly cloud-free composites with gap-filling across neighbouring months; Sentinel-1 radar as an optional back-up.
⚠ **Drought and nutrient shortage also make leaves yellow**	✓ Moisture indices (NDMI, MSI) plus rainfall and temperature normalisation (KSNDMC) isolate the disease-linked change.
⚠ **A 10 m pixel is larger than one palm**	✓ We map plot-level condition, verify with Google Earth Pro and GPS field points, and report “Suspected YLD” rather than a diagnosis.
⚠ **Coconut, banana and pepper grow among arecanut**	✓ Random Forest is trained on field-verified samples and learns each crop's month-by-month behaviour.
⚠ **The ≥ 10 m tree-height rule cannot be measured by Sentinel-2**	✓ Applied as a mature-stand mask built from field and reference data.

WHAT WE WILL DELIVER

- **Arecanut distribution maps** — Taluk-wise, Sringeri and Sulya
- **Health and YLD severity maps** — Healthy · Stressed · Suspected YLD
- **Multi-temporal report** — Trends, spread and hotspots
- **GIS-ready datasets** — GeoTIFF / GeoPackage for planning

WHO BENEFITS

- **Growers** — Early warning, so action can start before yield falls
- **Extension officers** — Target field visits and advice where stress is rising
- **Researchers** — Spread and hotspot data for disease-management studies
- **Planners** — Area-wise evidence for support and management planning

QUALITY CHECK Independent field and Google Earth test points · confusion matrix, overall accuracy, Kappa, F1 · targets: OA ≥ 85 %, Kappa ≥ 0.80 · expert review of suspected-YLD gardens